

CA3010: Review code for XAML injection vulnerabilities (code analysis)

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Content

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CA3010: Review code for XAML injection vulnerabilities

Summarize this article for me

Property Value	Rule ID CA3010	Title Review code for XAML injection vulnerabilities	Category Security	Fix is breaking or non-breaking Non-breaking	Enabled by default in .NET 10 No	Applicable languages C# and Visual Basic
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Cause

Potentially untrusted HTTP request input reaches a [System.Windows.Markup.XamlReader](#) Load method.

By default, this rule analyzes the entire codebase, but this is [configurable]().

Rule description

When working with untrusted input, be mindful of XAML injection attacks. XAML is a markup language that directly represents object instantiation and execution. That means elements created in XAML can interact with system resources (for example, network

access and file system IO). If an attacker can control the input to a [System.Windows.Markup.XamlReader](#) Load method call, then the attacker can execute code.

This rule attempts to find input from HTTP requests that reaches a [System.Windows.Markup.XamlReader](#) Load method.

Note

This rule can't track data across assemblies. For example, if one assembly reads the HTTP request input and then passes it to another assembly that loads XAML, this rule won't produce a warning.

Note

There is a configurable limit to how deep this rule will analyze data flow across method calls. See [Analyzer Configuration](#) for how to configure the limit in an EditorConfig file.

How to fix violations

Don't load untrusted XAML.

When to suppress warnings

Don't suppress warnings from this rule.

Configure code to analyze

Use the following options to configure which parts of your codebase to run this rule on.

- [Exclude specific symbols]()
- [Exclude specific types and their derived types]()

You can configure these options for just this rule, for all rules they apply to, or for all rules in this category ([Security](#)) that they apply to. For more information, see [Code quality rule configuration options](#).

Exclude specific symbols

You can exclude specific symbols, such as types and methods, from analysis by setting the [excluded_symbol_names](#) option. For example, to specify that the rule should not run on any code within types named `MyType`, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Symbol name only (includes all symbols with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#). Each symbol name requires a symbol-kind prefix, such as `M:` for methods, `T:` for types, and `N:` for namespaces.
- `.ctor` for constructors and `.cctor` for static constructors.

Examples:

Option Value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType</code>	Matches all symbols named <code>MyType</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType1 MyType2</code>	Matches all symbols named either <code>MyType1</code> or <code>MyType2</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS.MyType.MyMethod(ParamType)</code>	Matches specific method <code>MyMethod</code> with the specified fully qualified signature.
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS1.MyType1.MyMethod1(ParamType) M:NS2.MyType2.MyMethod2(ParamType)</code>	Matches specific methods <code>MyMethod1</code> and <code>MyMethod2</code> with the respective fully qualified signatures.

Exclude specific types and their derived types

You can exclude specific types and their derived types from analysis by setting the [exclude_d_type_names_with_derived_types](#) option. For example, to specify that the rule should not run on any methods within types named `MyType` and their derived types, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Type name only (includes all types with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#), with an optional `T:` prefix.

Examples:

Option value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType</code>	Matches all types named <code>MyType</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType1 MyType2</code>	Matches all types named either <code>MyType1</code> or <code>MyType2</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS.MyType</code>	Matches specific type <code>MyType</code> with given fully qualified name and all of its derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types =</code>	

M:NS1.MyType1|M:NS2.MyType2 | Matches specific types MyType1 and MyType2 with the respective fully qualified names, and all of their derived types. | |

Pseudo-code examples

Violation

```
using System;
using System.IO;

public partial class WebForm : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {
        string input = Request.Form["in"];
        byte[] bytes = Convert.FromBase64String(input);
        MemoryStream ms = new MemoryStream(bytes);
        System.Windows.Markup.XamlReader.Load(ms);
    }
}
```

```
Imports System
Imports System.IO

Public Partial Class WebForm
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(sender As Object, e As EventArgs)
        Dim input As String = Request.Form("in")
        Dim bytes As Byte() = Convert.FromBase64String(input)
        Dim ms As MemoryStream = New MemoryStream(bytes)
        System.Windows.Markup.XamlReader.Load(ms)
    End Sub
End Class
```